

Mid-Term Examination — Answer Key

Corresponds to: Sample Paper (Subject: General Science & Aptitude)

Section A — Multiple Choice

1. B
2. C
3. C
4. C

Section B — Short Answer (model answers, key points)

- i. Photosynthesis: plants convert light energy, water, and carbon dioxide into glucose and oxygen using chlorophyll.
- ii. Newton's Second Law: Force equals mass times acceleration ($F = m \times a$).
- iii. Speed is a scalar (magnitude only); velocity is a vector (magnitude and direction).

Section C — Weighted Long Answer (grading notes)

- a) Water cycle: evaporation, condensation, precipitation, collection — full marks require all four stages explained.
- b) Kinetic energy: $KE = \frac{1}{2} \times m \times v^2$, derived from work-energy theorem.
- c) Plant cell: cell wall, chloroplast, large central vacuole, nucleus, cytoplasm — award marks per labeled structure.